Mahmoud ElSalakawy

Senior Data Scientist | AI & ML Engineer

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SUMMARY

Data scientist with an MSc in Data Science (Cairo University, 2026) and 9+ years of telecom RAN engineering. Designed the Prophet-based anomaly-detection method behind AIOS, a tool in production for RAN teams in the UK and Egypt, and published the method as first author in a peer-reviewed journal. Rebuilt a production demand-forecasting pipeline as a freelance data scientist, and build bilingual Arabic/English RAG applications with MLflow, DVC and Docker.

CORE SKILLS

- **Data science and statistics:** time-series forecasting and anomaly detection (Prophet, SARIMAX, Isolation Forest), demand forecasting, feature engineering (cyclical and lag features), leakage-safe evaluation, hypothesis testing (Wilcoxon, Holm correction, effect sizes), balanced accuracy, MCC, macro-F1
- **Machine learning:** scikit-learn, XGBoost, LightGBM, CatBoost, TF-IDF text classification, SHAP explainability
- **Python and data:** Python, pandas, NumPy, ETL pipelines, Plotly, Power BI, SQLite
- **MLOps and engineering:** MLflow, DVC, Docker, Docker Compose, GitHub Actions, pytest, FastAPI, Pydantic, Flask, Streamlit, Git, GitHub
- **GenAI:** RAG, embeddings (Sentence Transformers, multilingual-e5), FAISS vector search, LLM APIs (OpenAI), function calling, grounding and citation validation
- **Cloud:** Google Cloud Certified Associate Cloud Engineer
- **Languages:** Arabic (native), English (professional)

EXPERIENCE

Solution Architect, AI/Automation Program, VOIS Egypt — 2025 – Present

Formally assigned role, about 20% of working time, alongside the RAN Integration role below.

- Designed the anomaly-detection method behind AIOS (Expert-Tuned Prophet), which reached 98.8% balanced accuracy and 0.984 MCC on 168 operator KPI series in offline research evaluation, and delivered it as a production tool for UK and Egypt RAN teams of a European mobile operator group.
- Built per-cell, per-KPI Prophet forecasting with KPI-specific direction and threshold rules and consecutive-anomaly logic; the UK configuration holds 73 KPI detection-rule entries across 2G, 3G, 4G and 5G.
- Separated the reusable detection logic from market-specific KPI schemas and thresholds, so the UK (Ericsson) and Egypt (Huawei and Ericsson) production setups and a Germany proof of concept used configuration changes, not code changes.
- Packaged AIOS as a standalone desktop application with local processing, SQLite persistence and Plotly and Excel reports, so engineers need no Python setup and network data is not sent to an external AI service (Python, Prophet, Flask, pandas).
- Presented AIOS to senior stakeholders in 2025 as a reusable platform and team capability; it was approved, and reuse in other new markets is being explored.
- Built a hackathon prototype (NPD&BE AI Hackathon, VOIS Egypt) that correlates KPI anomaly evidence with environmental alarms to produce structured root-cause findings and recommended checks, keeping engineers responsible for the final diagnosis (Python, RAG, FAISS, LLM API, Streamlit).
- Deliver two AI/GenAI awareness sessions to VOIS Egypt engineers (2025 and 2026).

Data Scientist (Freelance, remote), CATRION — Nov 2024 – Mar 2025

- As the sole data scientist, fixed data leakage and engineered cyclical and lag features in a 41.5M-meals-a-year demand forecasting pipeline, then redesigned the Power BI reports used for decisions.
- Rebuilt the ETL and model layer of the production forecasting pipeline.
- Presented findings to catering and supply-chain teams, turning technical metrics into waste-reduction recommendations.

Radio Expert, RAN Integration, VOIS Egypt — Mar 2018 – Present

Joined VOIS as a RAN Field Engineer in March 2018.

- Integrate 2G–5G radio networks across Ericsson and Huawei vendors.
- Started and led a five-track L&D program for about 70 Integration engineers, beginning with a needs survey (about 49 responses, ~70% participation); the program has delivered 8 sessions so far. Program lead since March 2026.
- Mentor peers through the program: designed a seven-session applied AI learning path, delivered the Statistics Foundations for Machine Learning session, and set up instructor briefs, quizzes and feedback surveys for volunteer instructors.

Radio Integration Engineer, Huawei — Apr 2017 – Mar 2018

- Worked with the Automation team to upgrade and develop Python tools.
- Provided field technical support for hardware, radio and performance issues, working with teams across multiple countries.

RESEARCH AND PUBLICATION

MSc Thesis: Enhancing Anomaly Detection in Mobile Network Using Hybrid Model, Cairo University 2026

- Compared four anomaly-detection methods (ETP, Prophet, SARIMAX, Prophet–Isolation Forest) using time-aware, leakage-safe evaluation and paired Wilcoxon tests with Holm correction ($p \leq 5.08 \times 10^{-6}$, RBC ≥ 0.93); published as first author in JICDS (2026).
- Documented a negative result: the Prophet–Isolation Forest hybrid did not improve on standalone Prophet, while expert-calibrated tolerance margins (ETP) gave significant gains.
- First author: "Policy-Calibrated Forecast Intervals for Anomaly Detection in Radio Access Mobile Network", Journal of Intelligent Computing and Data Science (JICDS), Cairo University, April 2026. DOI 10.21608/jicds.2026.457039.1026.

PROJECTS

Egyptian Civil Code RAG: Bilingual Arabic/English Legal Assistant, Personal MVP 2026

MLOps Practitioner graduation project. github.com/MahmoudAmgad88/civil-code-rag-mlops

- Built the data pipeline for a legal RAG system: PDF extraction, Arabic/English normalization, corpus validation and lineage tracking across 162 legal records and 268 retrieval records, with Recall@K and MRR evaluation tracked in MLflow.
- Answers Arabic and English questions with article-level citations, using multilingual-e5-small embeddings, citation validation and out-of-scope detection (FastAPI, Pydantic, OpenAI API).
- Versioned corpus and index with DVC, ran 73 automated tests with pytest and GitHub Actions in an earlier validated run, and cut the CPU Docker image from about 18.5 GB to 1.8 GB.

Ticket Classification ML Pipeline, Technical assessment

- Compared Logistic Regression, calibrated Linear SVC and Naive Bayes on 47,837 imbalanced tickets (largest class 7.7× the smallest), selecting on macro-F1; the chosen model reached 86.43% test macro-F1, 8.25 points above Naive Bayes and 1.22 above Logistic Regression.

EDUCATION

M.Sc. in Data Sciences, Faculty of Graduate Studies for Statistical Research, Cairo University Awarded Jul 2026

GPA 3.87. Courses include Time Series Forecasting, Advanced Machine Learning, Pattern Recognition and Advanced Bayesian Inference.

B.Sc. in Communications & Electronics Engineering, Cairo University 2011

CERTIFICATIONS

- Google Cloud Certified Associate Cloud Engineer, July 2026
- Certified SAFe 6 Practitioner, valid until February 2027
- MLOps Practitioner (MLOps Crash Course), 2026